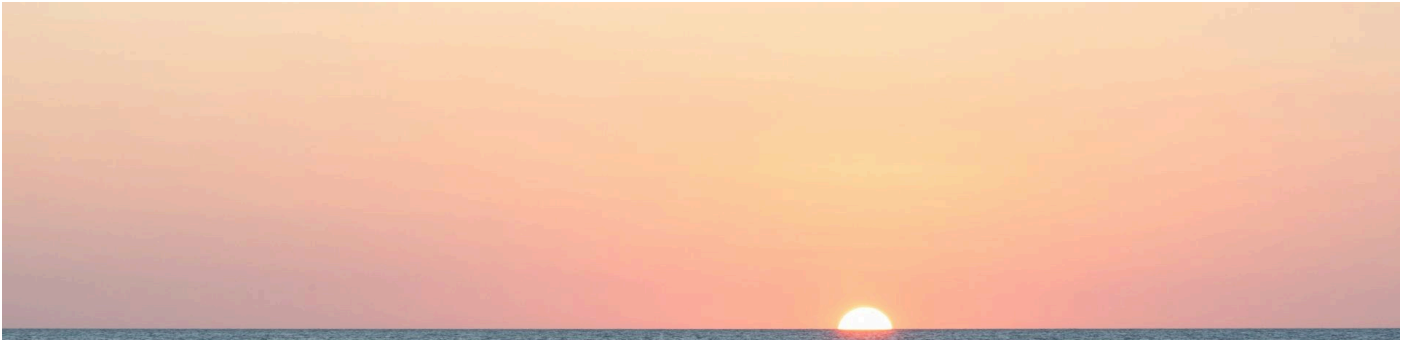


**52 | OVERVIEW**


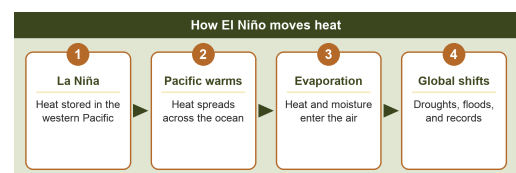
# Understanding ENSO: El Niño, Climate Health, and Global Weather Extremes

## Abstract

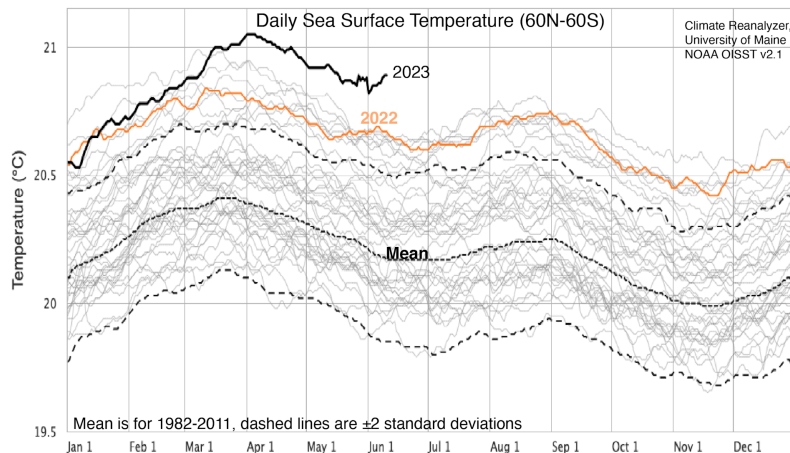
In June 2023, NOAA declared that El Niño had arrived, ending three years of La Niña. Climate scientist Kevin Trenberth explained how the tropical Pacific's ocean and atmosphere interact, what El Niño means for New Zealand and the world, and why global warming makes all of El Niño's impacts, from droughts to floods, more severe.

## Introduction

For three years, La Niña brought prevailing northeasterly winds to New Zealand and an unprecedented number of tropical and subtropical storms. In 2022, New Zealand had its highest annual mean temperature on record, and in early 2023 came a record rain event in Auckland followed three weeks later by Cyclone Gabrielle. When La Niña ended in March 2023, global sea surface temperatures suddenly hit record highs as the tropical Pacific warmed. El Niño was on its way. Trenberth compares the tropical climate to a roller coaster: La Niña at the bottom, El Niño at the top, occurring every three to seven years or so.



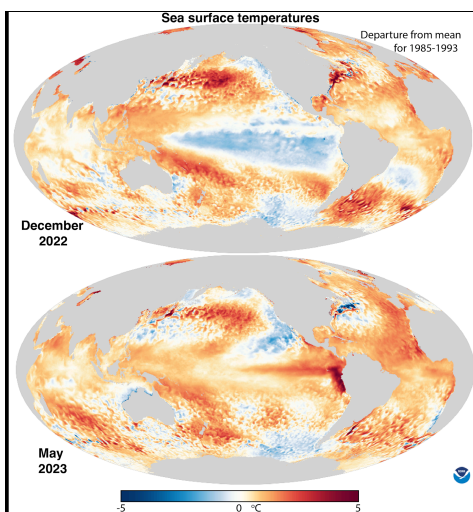
# How Do Scientists Study This?



## FROM THE ARTICLE

Daily global sea surface temperatures (60°N-60°S), showing 2023's record highs compared with other years. Image: University of Maine Climate Reanalyzer, author provided.

Scientists track ENSO by comparing sea surface temperatures with long-term averages. Trenberth compared global sea surface temperatures in 2023 with every previous year on record, and compared maps of temperature departures from December 2022 and May 2023 to reveal the transformation across the tropical Pacific, including a coastal El Niño off Peru and Ecuador. He explained the physics of ocean-atmosphere coupling: surface winds drive ocean currents and sea temperatures, and sea temperatures in turn drive winds and storms. Heat stored in the western Pacific during La Niña moves into the atmosphere during El Niño through evaporation, shifting rainfall and setting up teleconnections to distant regions.



## FROM THE ARTICLE

Sea surface temperature departures from the mean, December 2022 compared with May 2023. Image: NOAA, author provided.

# Explore the Evidence

## How El Niño moves heat

1

### La Niña

Heat stored in the western Pacific

2

### Pacific warms

Heat spreads across the ocean

3

### Evaporation

Heat and moisture enter the air

4

### Global shifts

Droughts, floods, and records

### Compare the records

2023 sea surface temperatures were plotted against every previous year.

### Map the change

Temperature maps from December 2022 and May 2023 showed the Pacific warming.

### Trace the connections

Scientists followed how heat and rainfall shifts alter weather worldwide.

## What Did They Discover?

During El Niño, heat stored in the western Pacific moves into the atmosphere, cooling the ocean and moistening the air. This shifts rainfall and bends jet streams and storm tracks in both hemispheres. New Zealand is one of the few countries that tends to be cooler during El Niño, with more southwesterly winds, drier eastern areas, and more rain on the West Coast. Globally, El Niño is the largest cause of droughts, raising wildfire risk in Australia, Indonesia, and Brazil, while bringing heavier rain to Peru and Ecuador. Research shows El Niño persistently reduces economic growth, with damages in the trillions of dollars. The warmest years come late in El Niño events, and 2024 was expected to break records.

### ENSO by the numbers

**3–7 yrs**

typical time between El Niño events

**3**

super El Niños: 1982-83, 1997-98, 2015-16

**2022**

New Zealand's warmest year on record

There is little evidence that climate change has altered El Niño itself, but all of its impacts are made worse by global warming.

## Takeaways

### A coupled system

Pacific winds and sea temperatures drive each other, creating El Niño and La Niña.

### Global reach

El Niño shifts droughts, floods, and storms far beyond the Pacific.

### Warming amplifies

Global warming intensifies El Niño's floods, droughts, and heat.

# Key Terms

## ENSO

The El Niño-Southern Oscillation, the linked ocean and atmosphere cycle in the tropical Pacific.

## Southern Oscillation

The see-saw of air pressure across the tropical Pacific that accompanies El Niño and La Niña.

## Coupling

When two systems, such as ocean and atmosphere, strongly influence each other.

## Teleconnection

A link between weather in one region and conditions far away.

## Atmospheric river

A narrow band of moisture in the atmosphere that can bring torrential rain.

### Match-Up: draw a line from each term to its clue

**A** ENSO

**B** Southern Oscillation

**C** Coupling

**D** Teleconnection

**E** Atmospheric river

**1** A link between weather in one region and conditions far away

**2** A narrow band of moisture in the atmosphere that can bring torrential...

**3** The see-saw of air pressure across the tropical Pacific that accompanies El...

**4** The El Niño-Southern Oscillation, the linked ocean and atmosphere cycle in the...

**5** When two systems strongly influence each other

Answer key: A-4 · B-3 · C-5 · D-1 · E-2

## Activity

### The ENSO Roller Coaster

Students model ENSO as a roller coaster and predict regional weather during each phase.

Materials: Roller coaster diagram template; World map; Weather symbol stickers

1. Draw a roller coaster track. Label the bottom La Niña and the top El Niño.
2. Add notes explaining how often El Niño occurs and how long its most intense phase lasts.
3. On the world map, place weather symbols for El Niño: drought in Australia, Indonesia, and Brazil; rain in Peru and Ecuador.
4. Add New Zealand's typical El Niño weather: cooler, southwesterly winds, dry east, wet west coast.
5. Explain how global warming changes the height of the ride even if the roller coaster stays the same.

## References

Trenberth, K. (2023). El Niño combined with global warming means big changes for New Zealand's weather. The Conversation. <https://doi.org/10.64628/AA.vyevj6nsj>